RGB Protocol on Bitcoin

The Basics

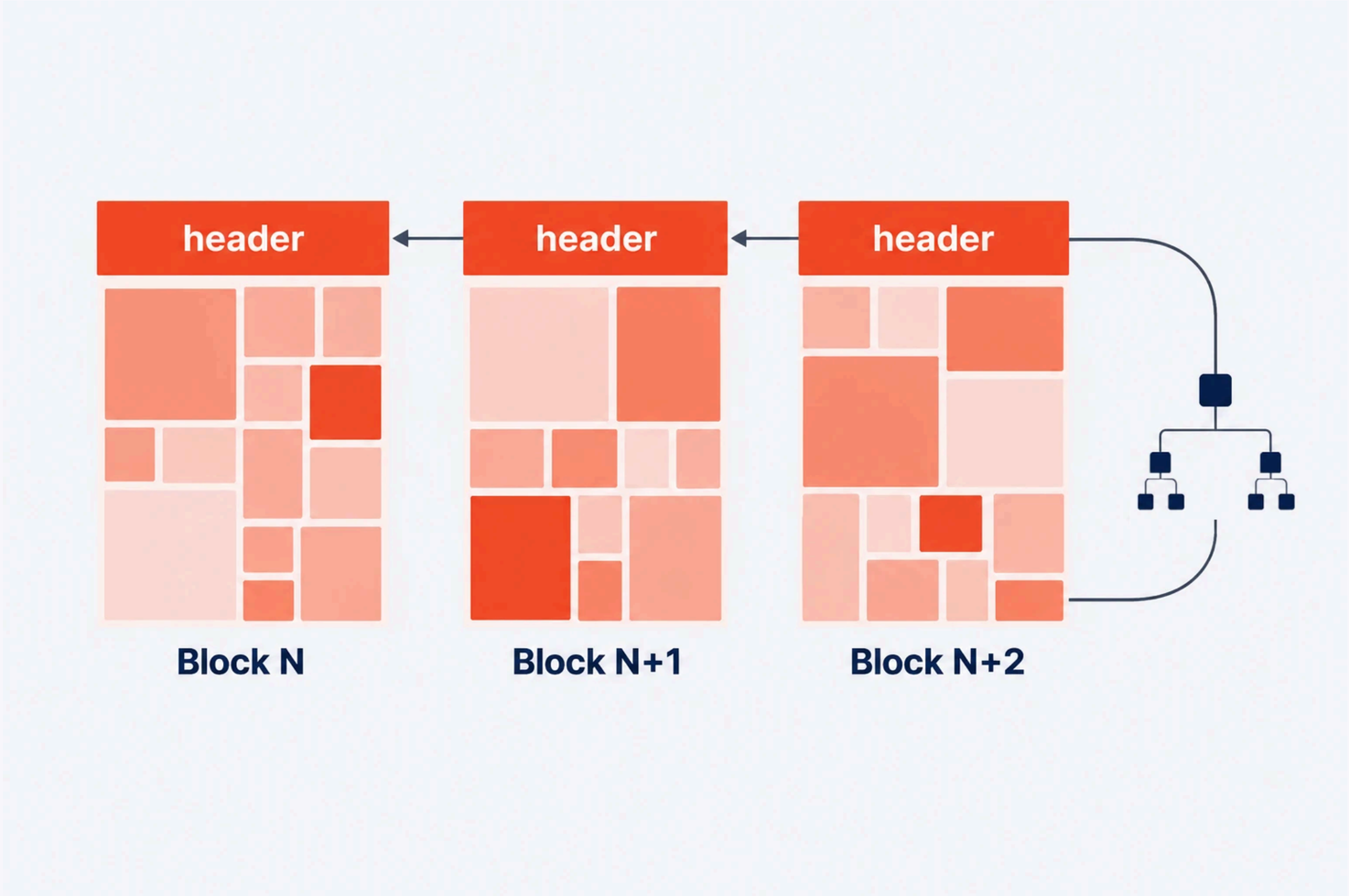
RGB Hub

Issuing and transferring assets

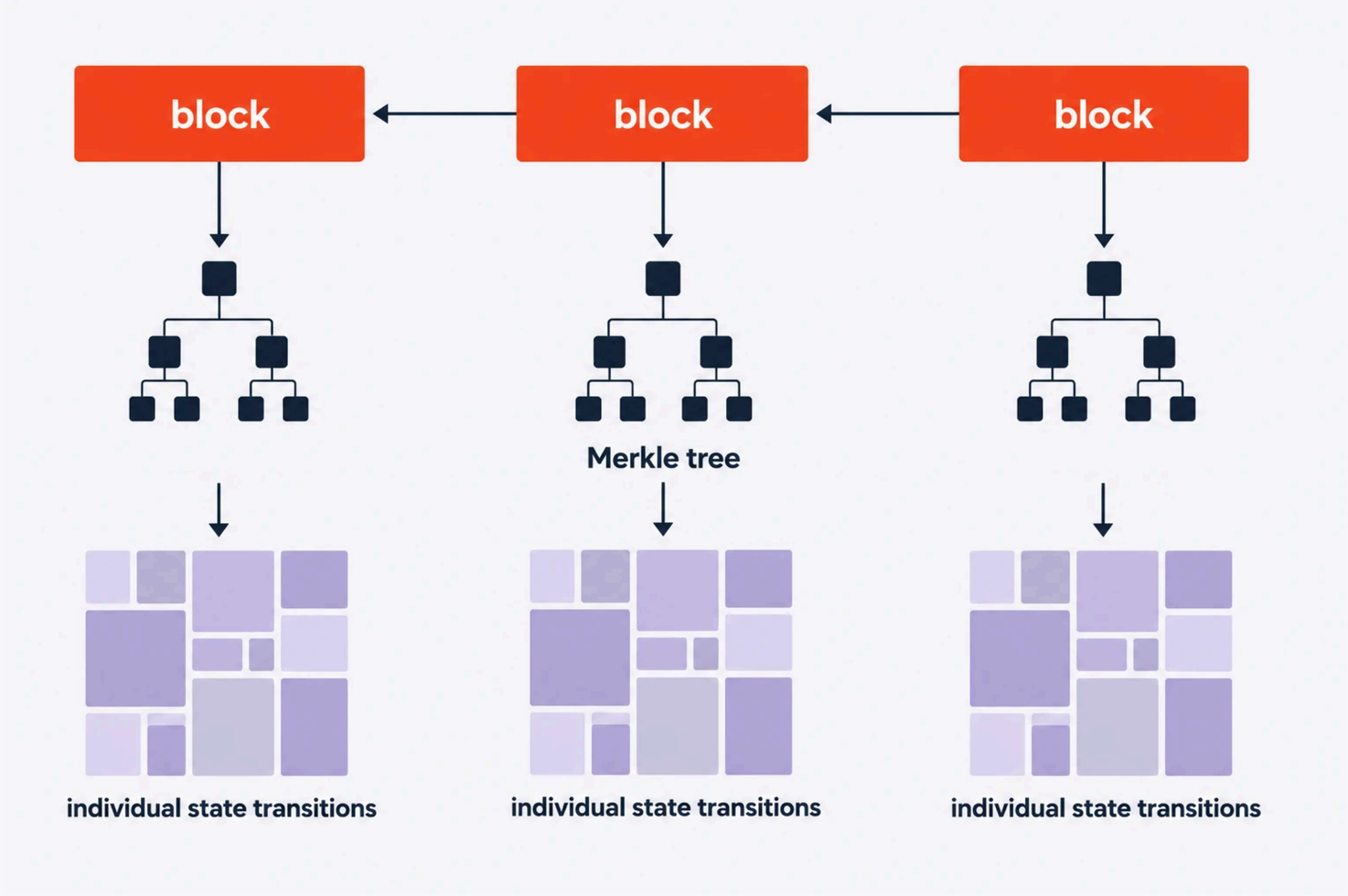
- Self-custody
- Privacy
- Scalability

RGB Hub

Why not put everything on the blockchain?



Only the parties involved



Client-side validation

Local validation

Only the parties involved verify the asset data.

Local balances

No block explorer shows RGB balances on chain.

Clueless public

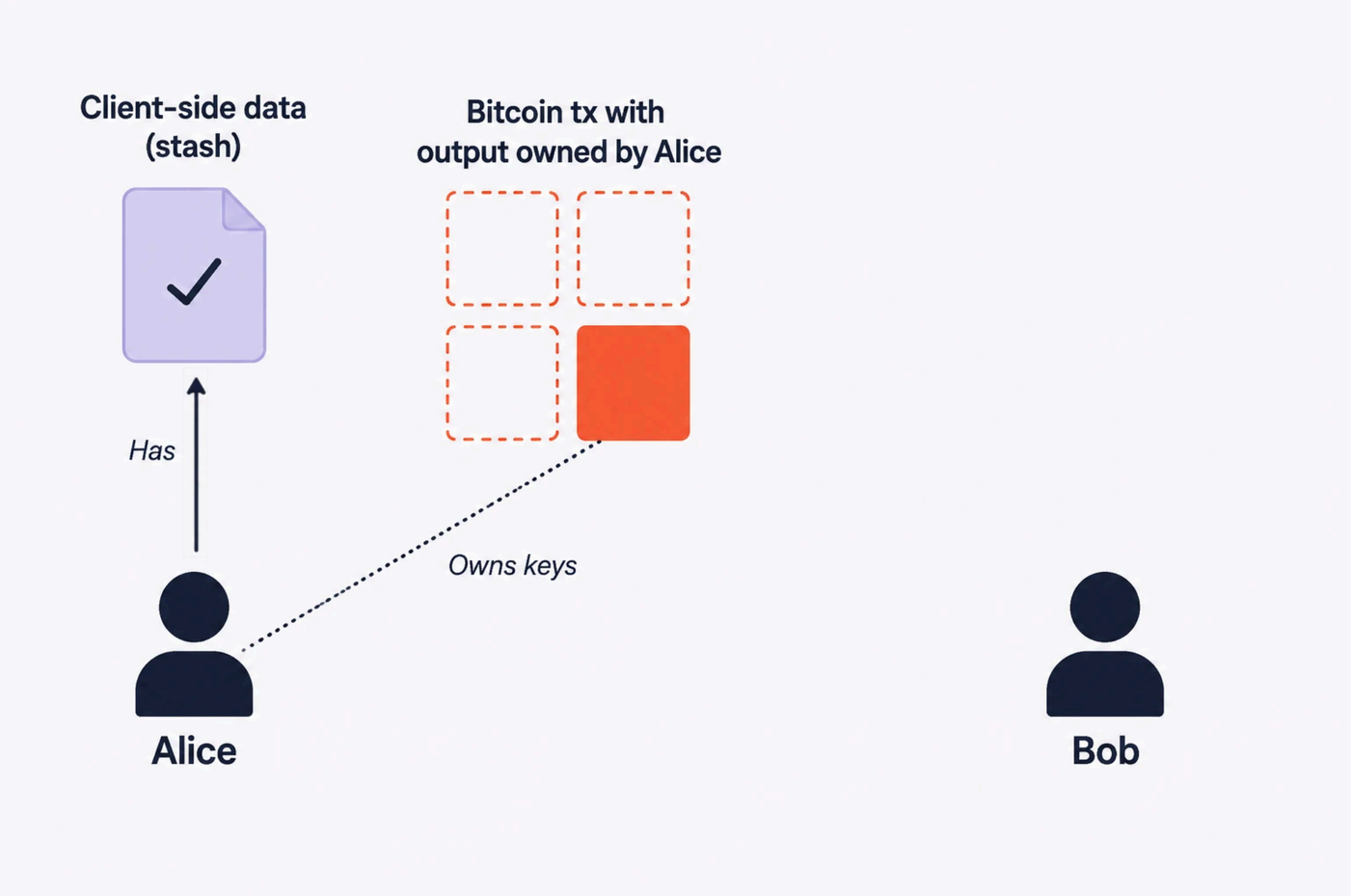
Everything the network sees is ordinary Bitcoin transactions.

A public commitment

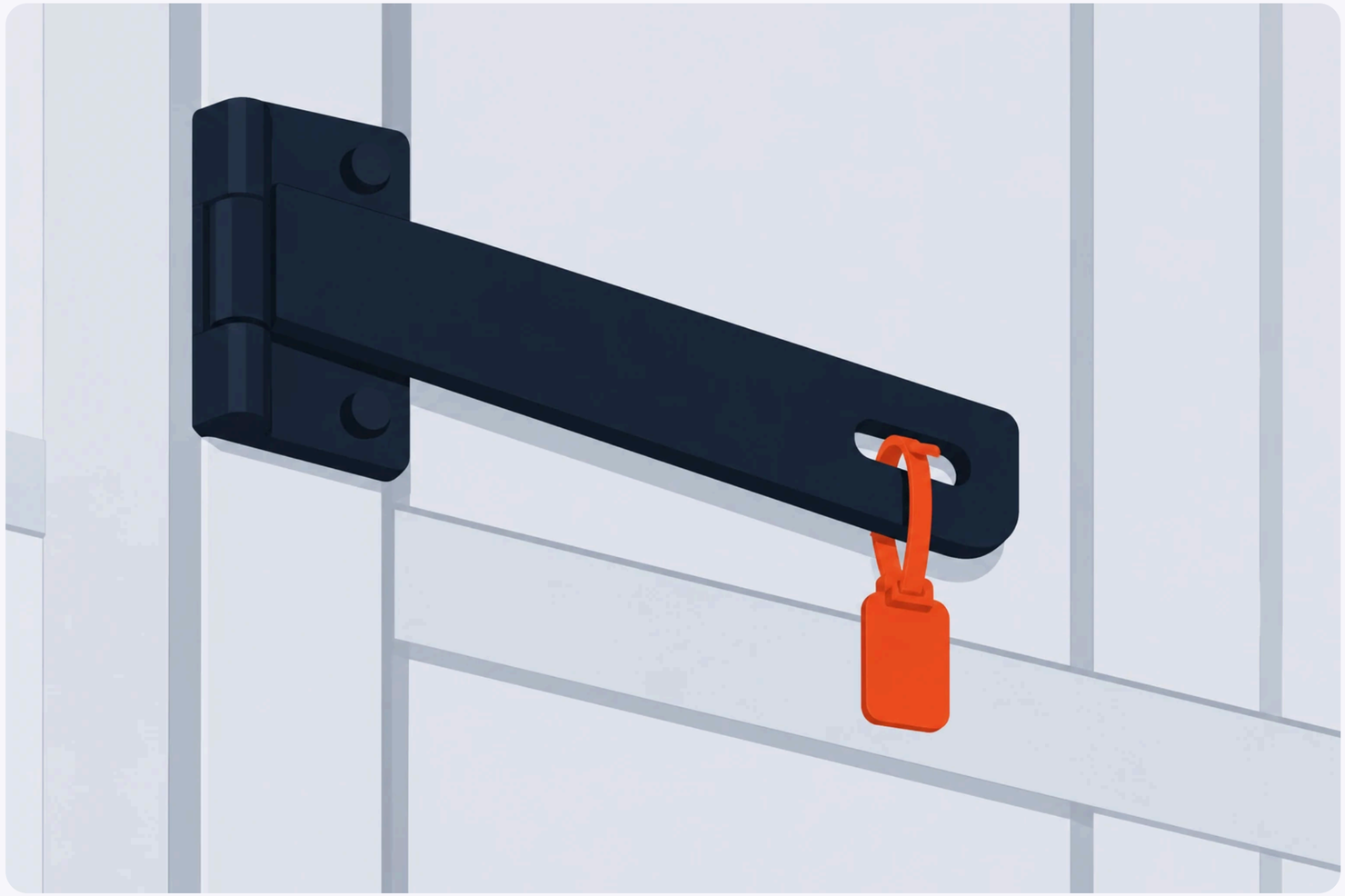
The blockchain only stores a cryptographic fingerprint.

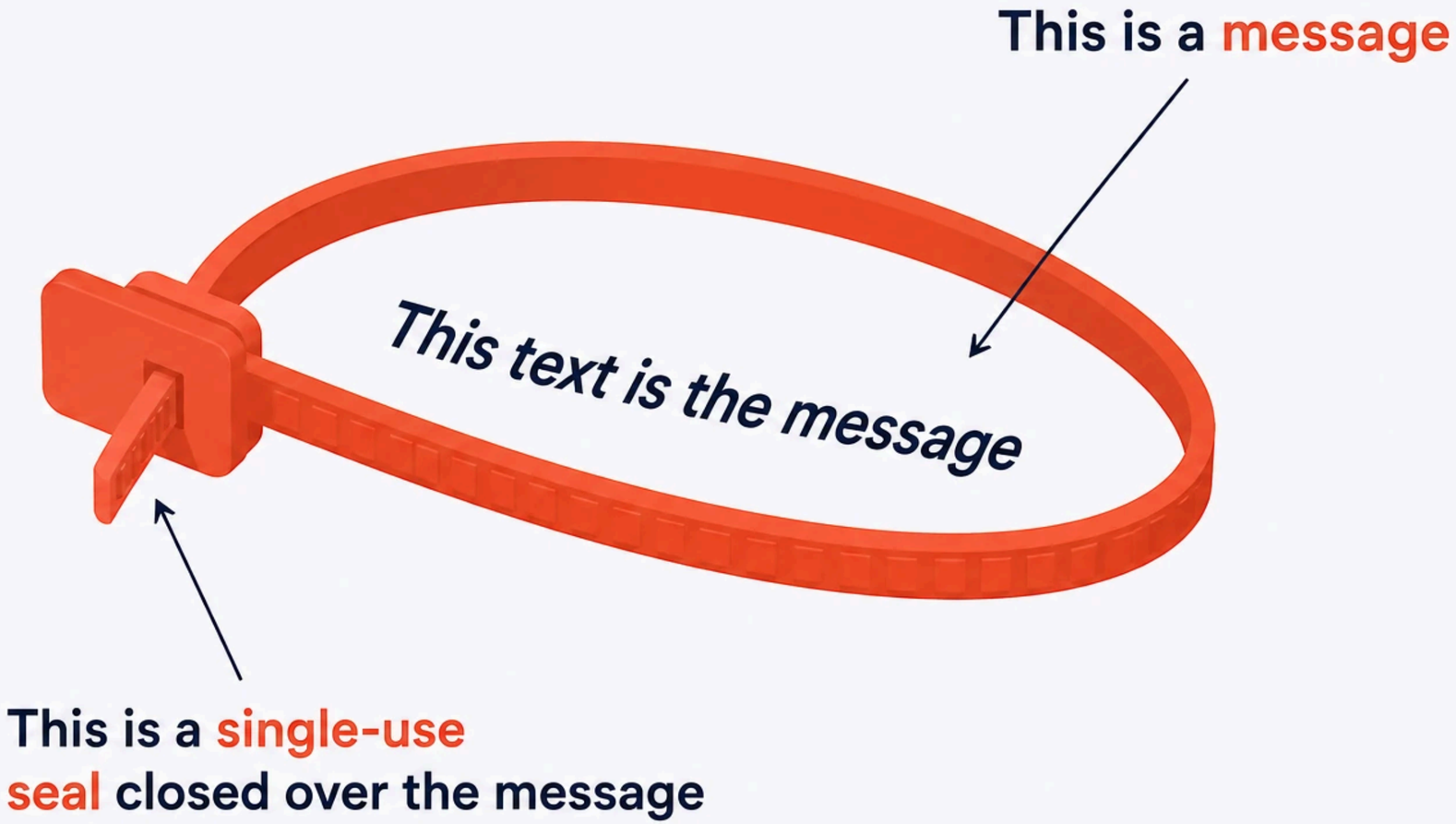
Where does an RGB asset live?

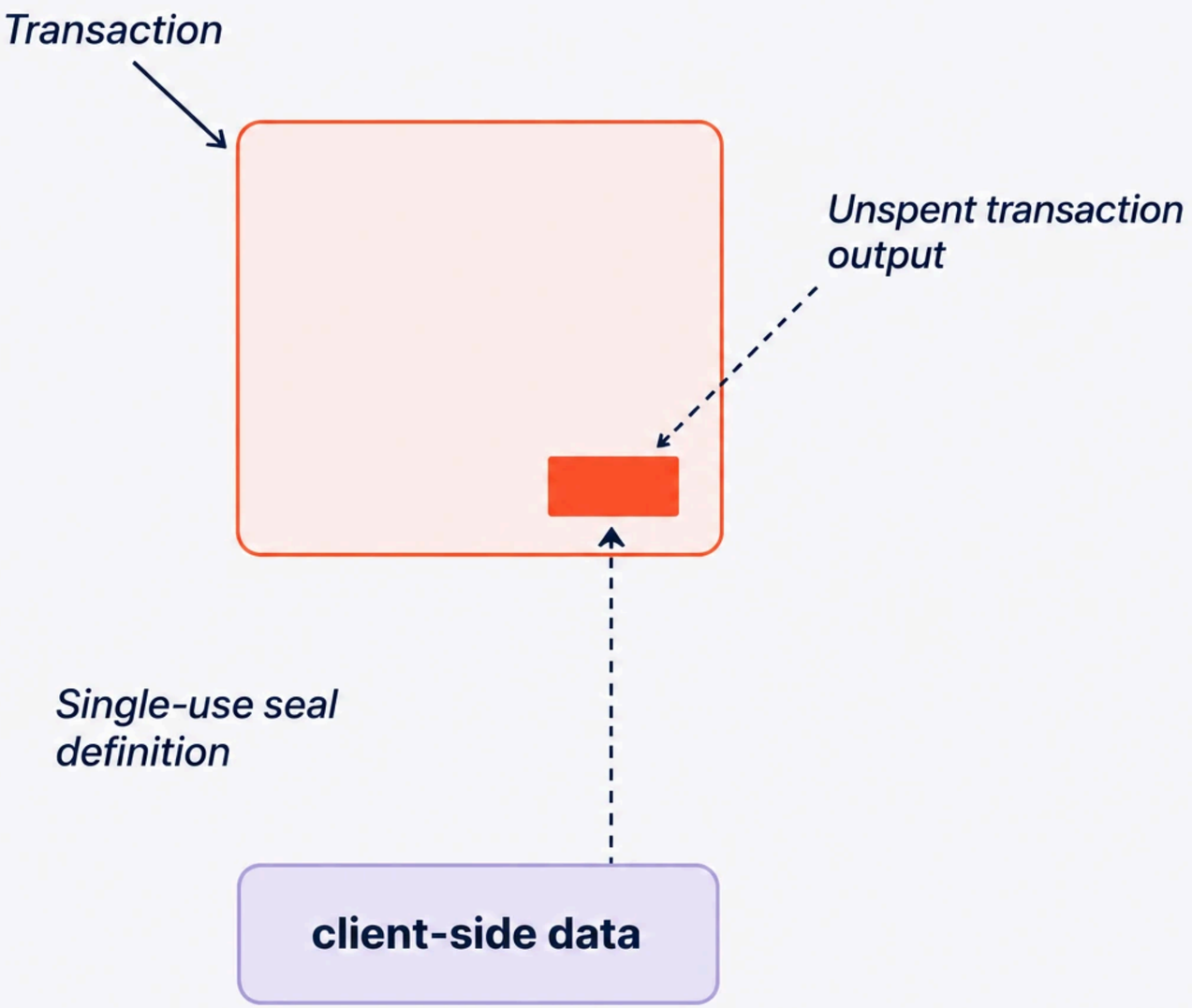
- The data
- The key(s)

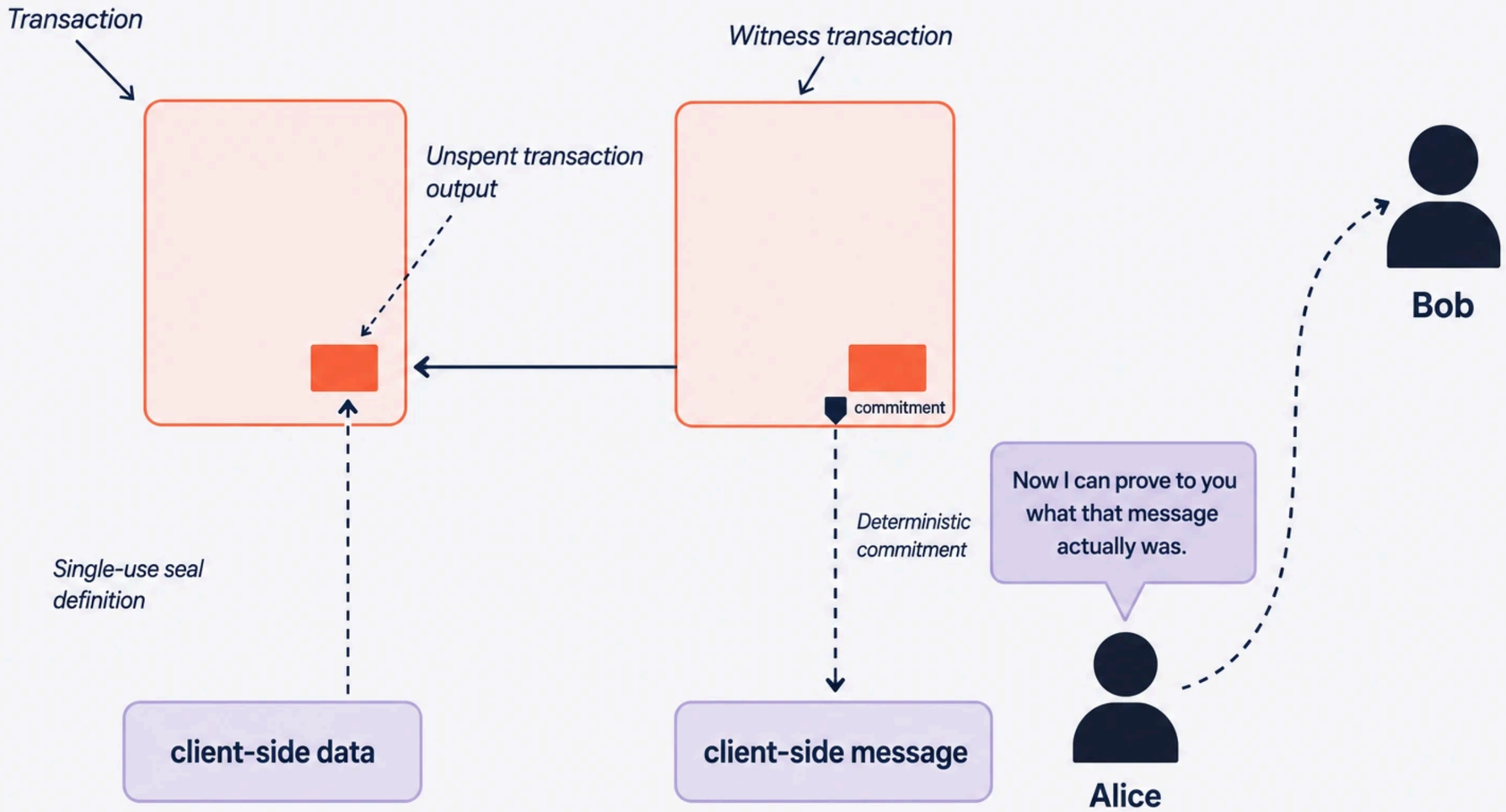


Single-use seals



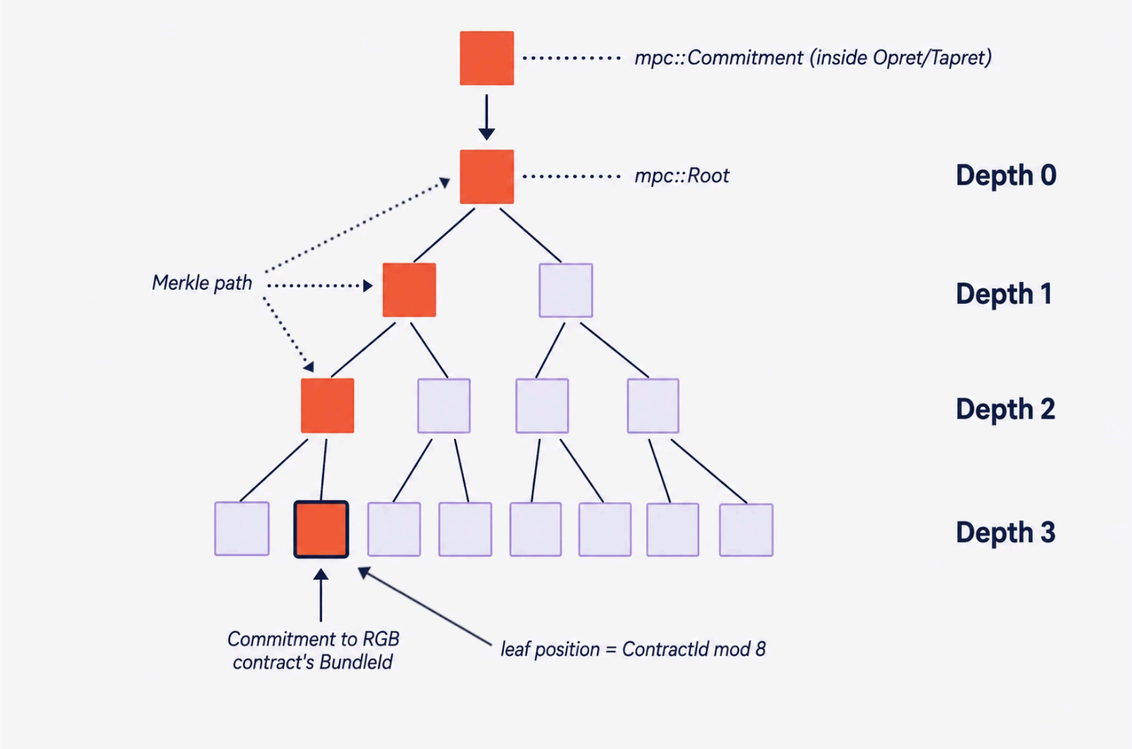

This is a message
This text is the message
This is a single-use
seal closed over the message


Transaction
Unspent transaction output
Single-use seal definition
client-side data

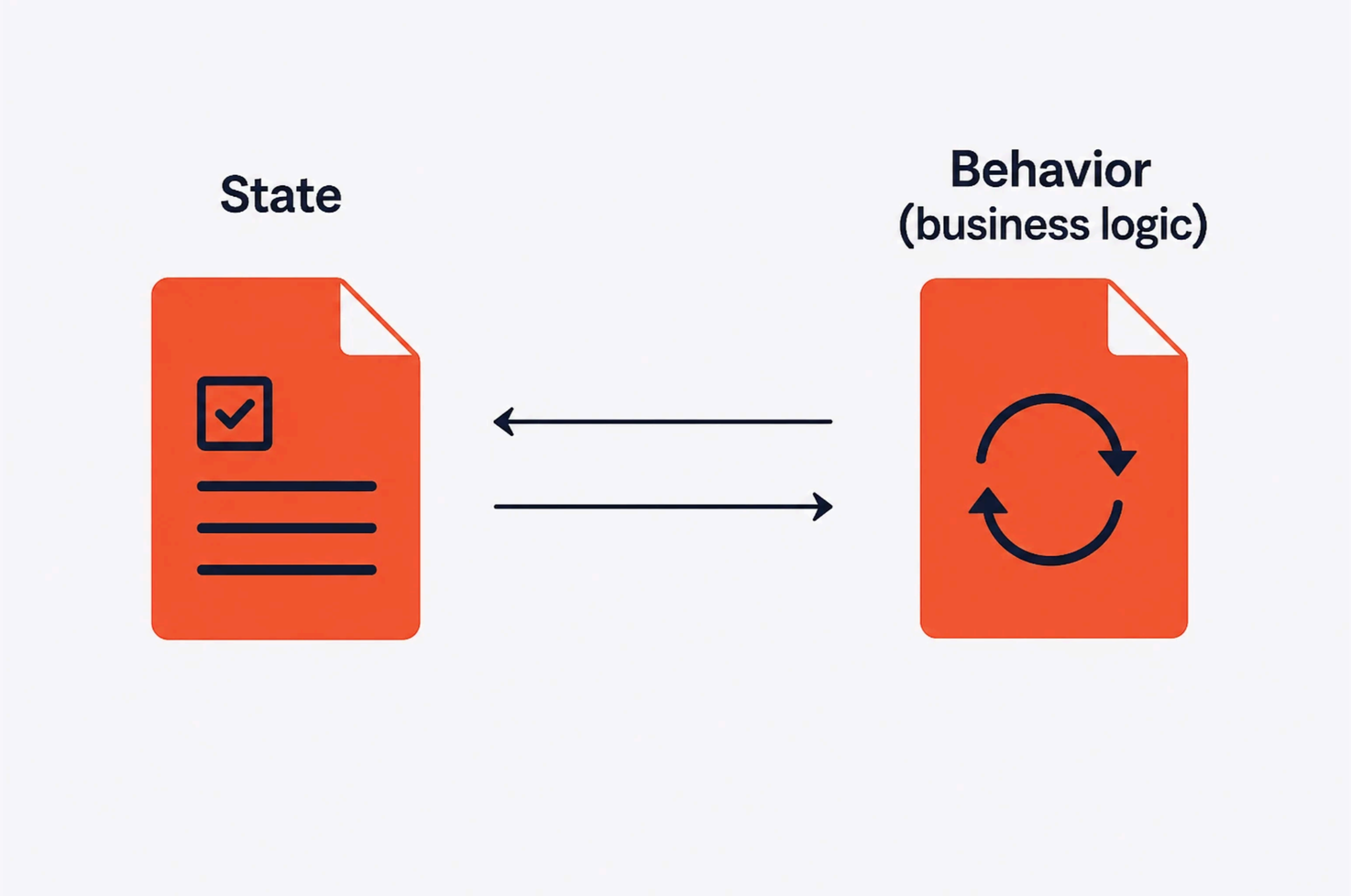

Transaction
Witness transaction
Unspent transaction output
commitment
Deterministic commitment
Single-use seal definition
client-side data
client-side message
Now I can prove to you what that message actually was.
Bob
Alice

Where is the commitment?

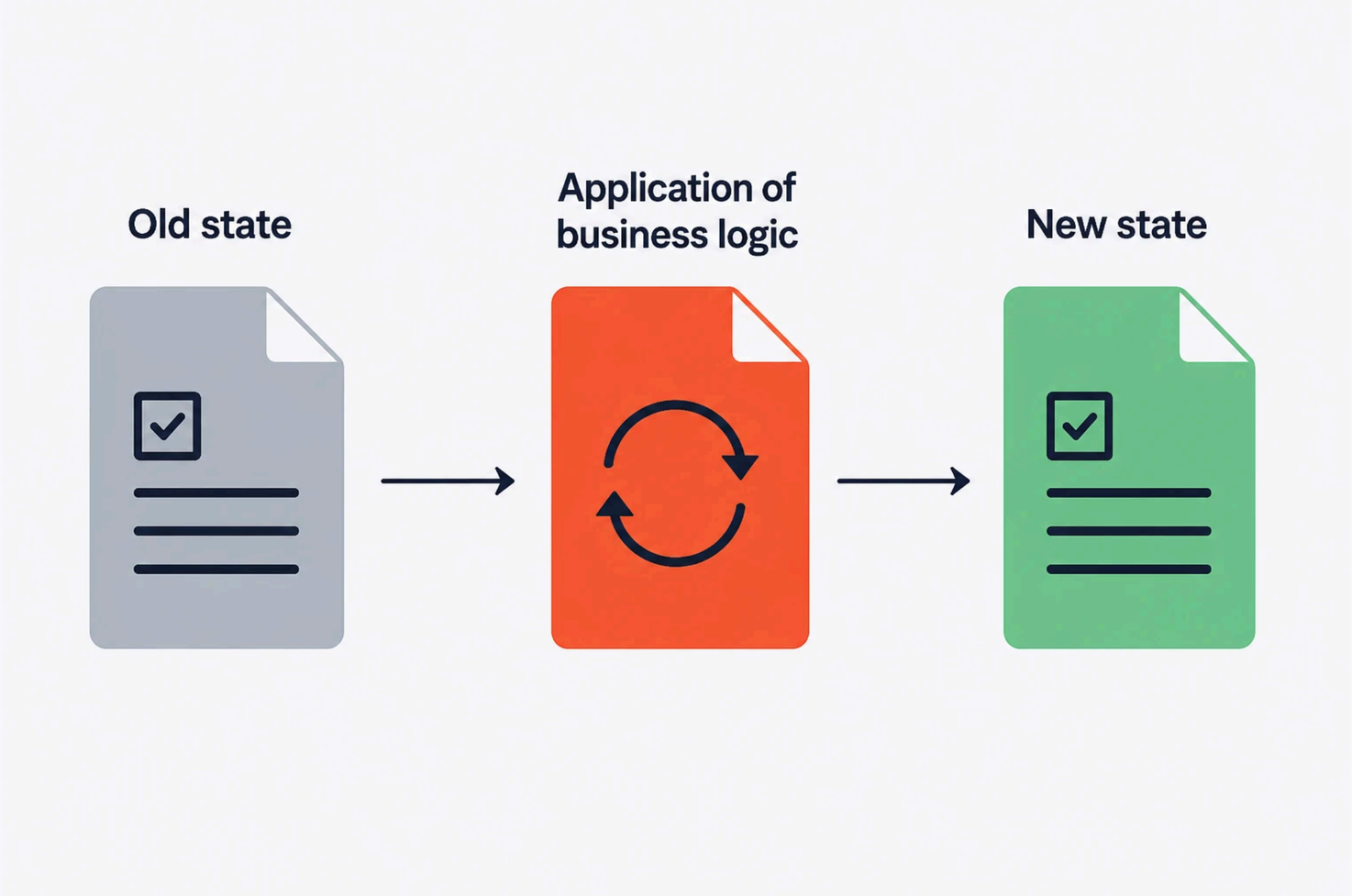
- Opret
- Tapret



Contracts, states and transitions



Contracts, states and transitions



Owned state and global state

OWNED STATE

Who has what

- Belongs to someone
- Attached to a seal: whoever controls the UTXO owns it

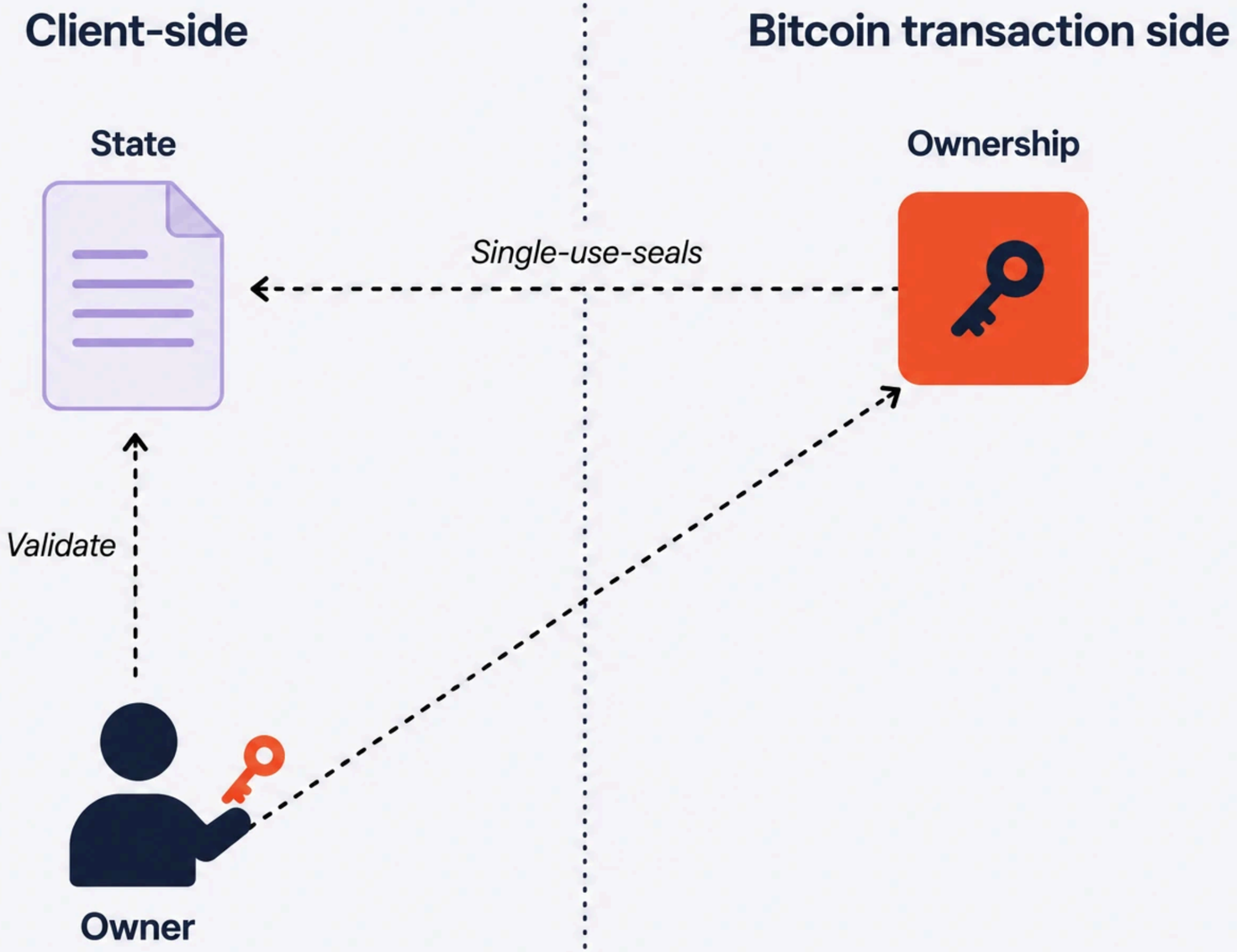
Example: **Alice's 500 dollars**

GLOBAL STATE

What the asset is

- Describes the contract
- Belongs to no one: everyone who holds the asset sees the same

Example: **name, ticker, total supply**


Client-side
Bitcoin transaction side
State
Ownership
Single-use-seals
Validate
Owner

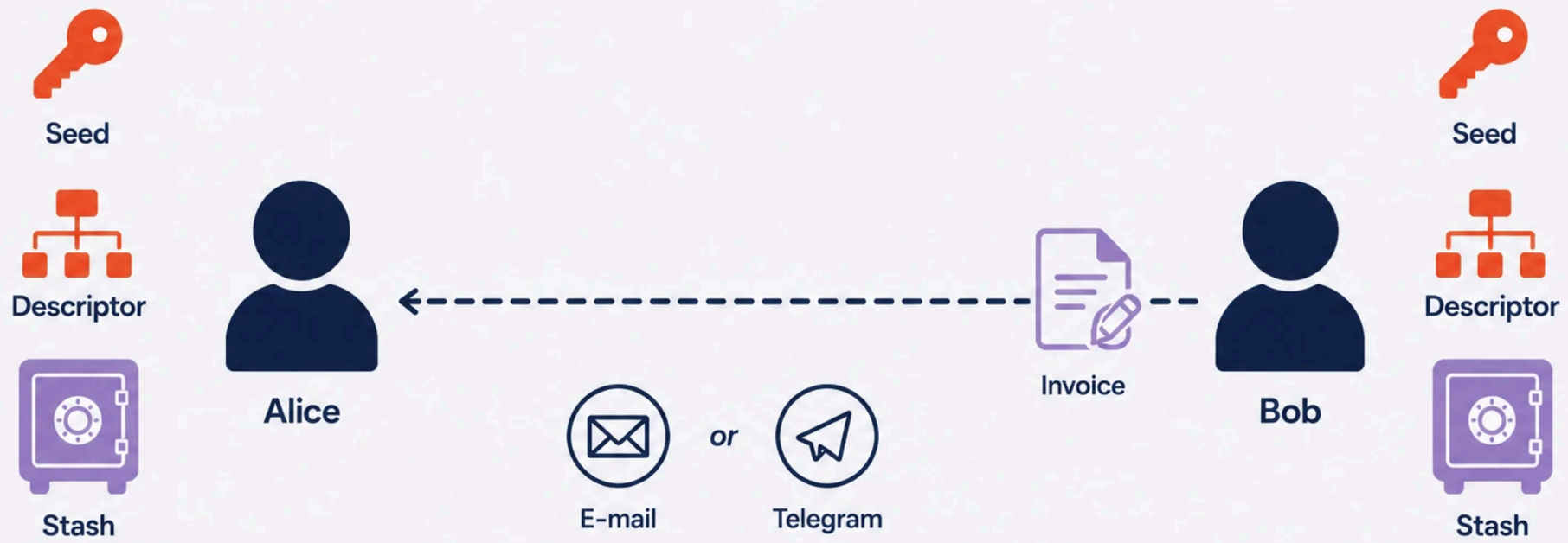
RGB schemas

A template that defines the contract's data and the operations it allows.

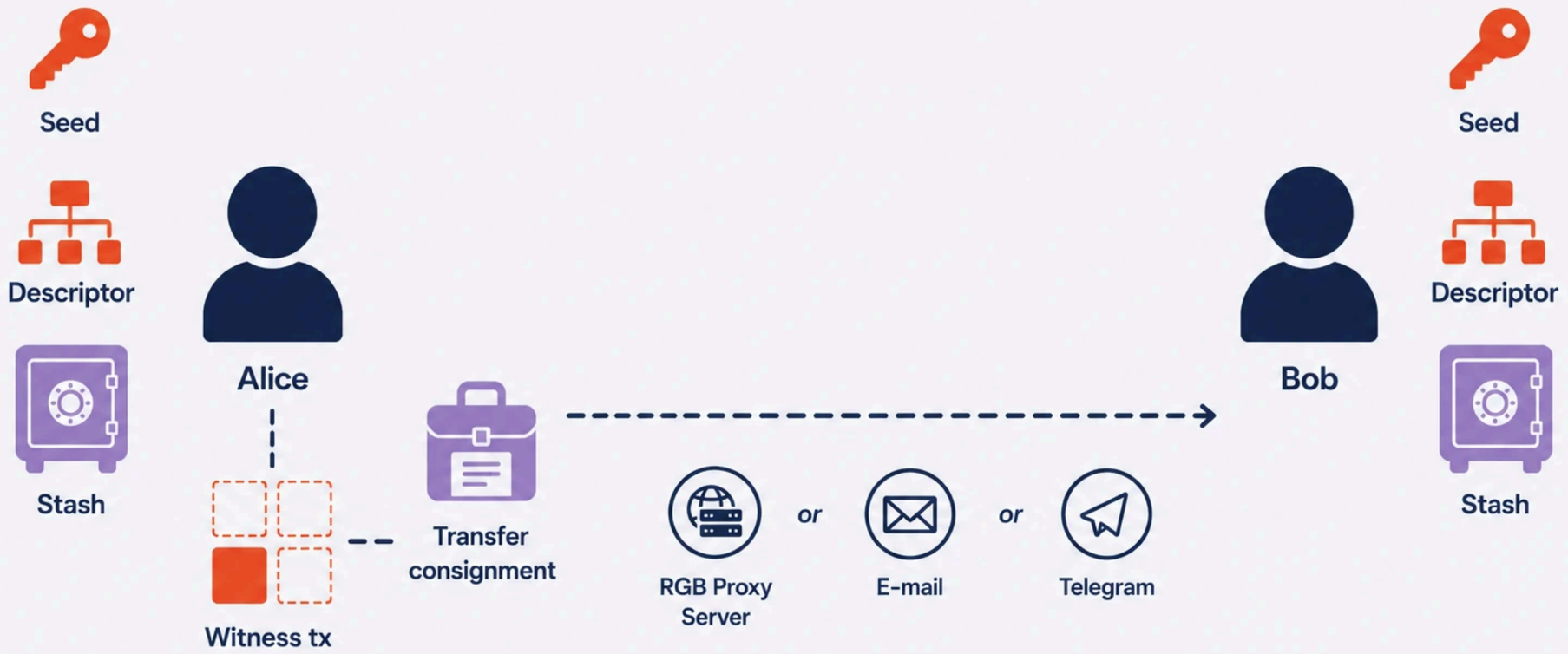
e.g. **NIA** or **IFA**

Supported schemas: IFA · docs.rgb.info

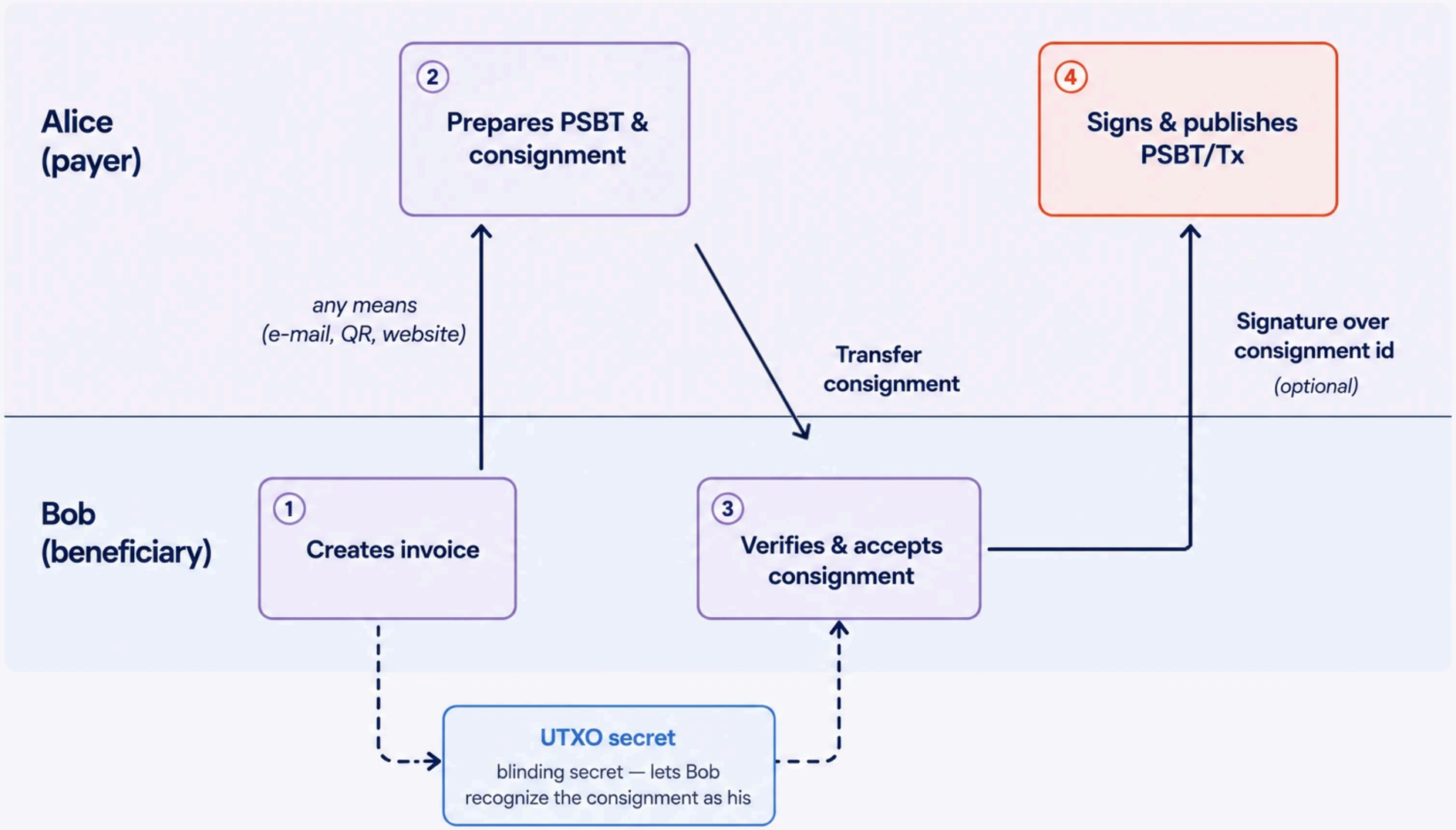
Bob prepares an invoice for Alice (seal definition, blinded or clear)



Alice prepares the witness transaction and sends the transfer consignment to Bob

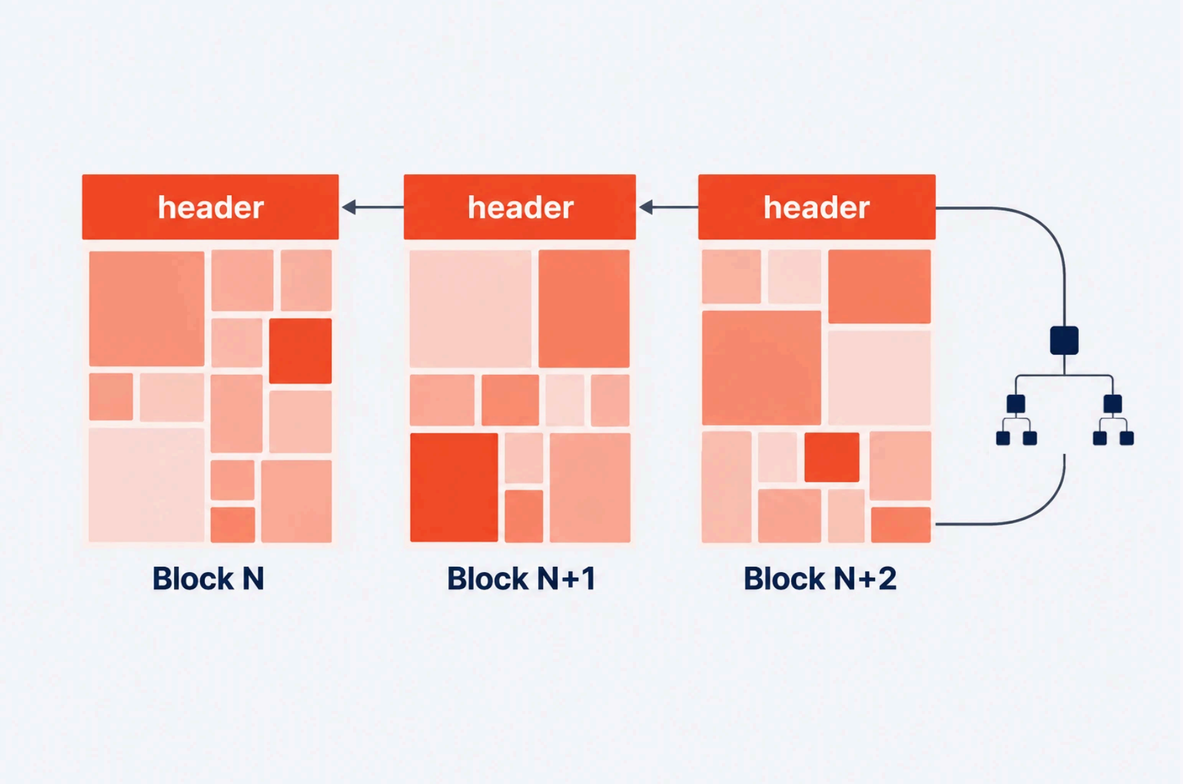


Transfer workflow — summary of the full exchange between Alice and Bob

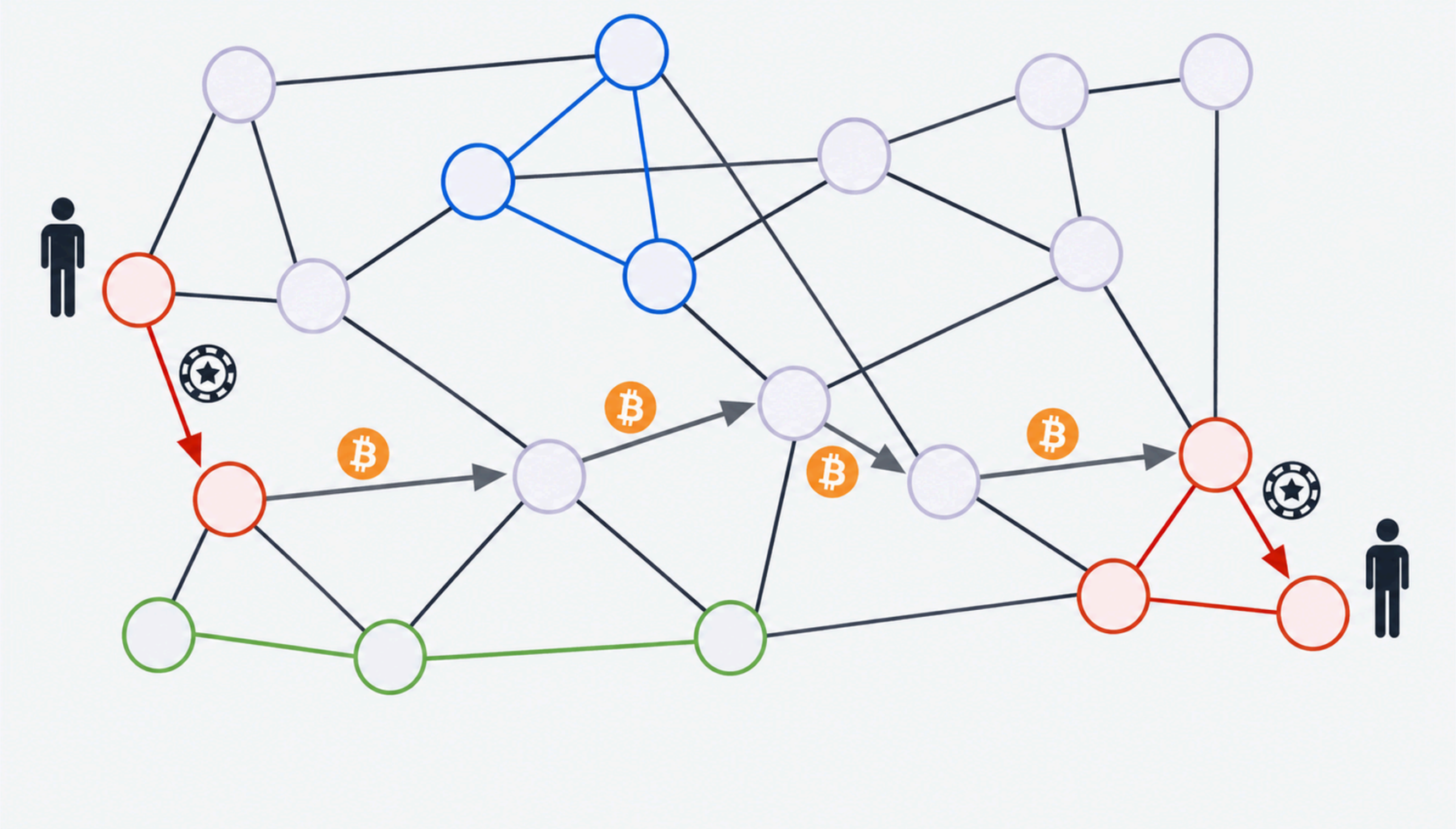


What an RGB wallet needs

- A Bitcoin wallet
- A way to read the blockchain
- A way to exchange consignments



RGB assets on the Lightning Network



Protocol documentation

rgb.info

docs.rgb.info

docs.rgb.info

The repositories

Protocol libraries

`github.com/rgb-protocol`

Tools to build

`github.com/RGB-Tools`

rgb-lib

- Rust library for building RGB wallets
- Bitcoin Development Kit
- Bindings for other languages

Language	Tool	Platform	Repository
C	c-ffi	Linux, macOS, Windows	`c-ffi`
C++	c-ffi	Linux, macOS, Windows	`c-ffi`
Kotlin	uniffi	Android	`rgb-lib-kotlin`
Node.js	c-ffi	Linux, macOS, Windows	`rgb-lib-nodejs`
Python	uniffi	Linux, macOS, Windows	`rgb-lib-python`
Swift	uniffi	iOS, macOS	`rgb-lib-swift`

Start your regtest

1 **Clone rgb-lib and start the services**

Source: `github.com/RGB-Tools/rgb-lib/tree/master/tests`

```
$ git clone https://github.com/RGB-Tools/rgb-lib
$ cd rgb-lib
$ ./tests/regtest.sh prepare_tests_environment
```

2 **While you test: test bitcoin, mining, stop**

All the commands are in the script:
`github.com/RGB-Tools/rgb-lib/tree/master/tests`

Requires Docker + Docker Compose plugin

Try the demo notebook

All the commands are in the README of the demo folder

github.com/RGB-Tools/rgb-lib-python/tree/master/demo

Stop your regtest first: the demo uses the same port.

Start

```
$ git clone https://github.com/
  RGB-Tools/rgb-lib-python
$ cd rgb-lib-python/demo
$ ./services.sh build # once
$ ./services.sh start
→ open the localhost:8888 link
```

While you run the cells

```
$ ./services.sh fund <address>
$ ./services.sh mine
$ ./services.sh stop # at the end
```

Run the cells in order with **Shift+Enter**.

Where to find functions

All rgb-lib functions (the “menu”)

github.com/RGB-Tools/rgb-lib/blob/master/bindings/uniffi/src/rgb-lib.udl

Transfer stuck? How transfers work, step by step

github.com/RGB-Tools/rgb-lib/tree/master/docs

Four things that work differently on RGB

One wallet, one place

Other apps don't know about your RGB assets.

The seed is not enough

You need to restore asset data, so back it up.

Prepare UTXOs

Create UTXOs before you can receive an asset.

On regtest, you mine

If a transfer seems stuck, mine one block and refresh both wallets.

Where WDK fits

WATCH-ONLY

rgb-lib

- RGB assets, invoices, transfers
- No private keys

transaction to sign

signed

SELF-CUSTODIAL WALLET

WDK

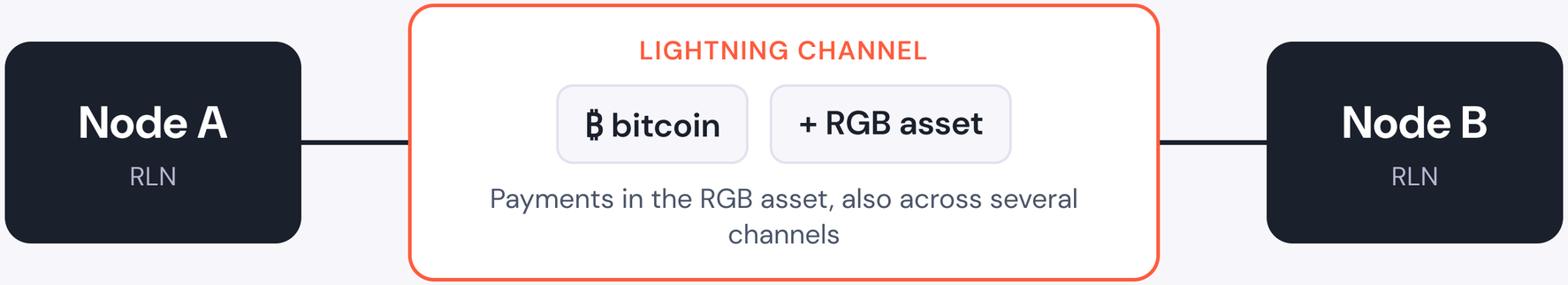
- Holds the keys
- Signs the transactions

Optional · earns bonus points

How to connect them: **see the WDK tutorial**

RGB Lightning Node (RLN)

A Lightning node based on LDK, extended to support RGB. Your app controls it through a REST JSON API.



USE RLN

Many small, fast payments, like an AI agent paying for things

RGB-LIB IS ENOUGH

Issuing an asset and making a few transfers

Run RLN locally

1. **Build it** (Rust: do it at home)

Source: github.com/RGB-Tools/rgb-lightning-node#install

```
$ git clone https://github.com/RGB-Tools/rgb-lightning-node --recurse-submodules --shallow-submodules
$ cd rgb-lightning-node && cargo install --locked --path .
```

2. **Start its regtest** (stop rgb-lib's environment first)

Source: github.com/RGB-Tools/rgb-lightning-node#regtest

```
$ ./regtest.sh start
```

3. **Start at least two nodes** (one shell each, own data folder and ports)

Source: github.com/RGB-Tools/rgb-lightning-node#regtest

```
$ rgb-lightning-node dataldk0/ --daemon-listening-port 3001 \
    --ldk-peer-listening-port 9735 --network regtest --disable-authentication
$ rgb-lightning-node dataldk1/ --daemon-listening-port 3002 \
    --ldk-peer-listening-port 9736 --network regtest --disable-authentication
```

4. **Initialize and unlock each node** through the API

Source: github.com/RGB-Tools/rgb-lightning-node#use

```
POST /init  → then  POST /unlock  # local services: see README, "Regtest" section
```


README: github.com/RGB-Tools/rgb-lightning-node
API (Swagger UI): rgb-tools.github.io/rgb-lightning-node

Your checklist

BEFORE SATURDAY

- ✓ **Install Docker** + the Compose plugin
- ✓ **Run your regtest once at home**
  The first start downloads the Docker images
- ✓ **Using Lightning?** Build RLN at home
- ✓ **Use RGB-Tools and rgb-protocol only**, v0.11.1
  Old repos like `rust-rgb20`, `rgb-node`, `rgb-lightning-sample` are archived

WITH DOCKER

- ! **One environment at a time**
  The demo, rgb-lib and RLN all use port 3000
- ! **Linux:** add your user to the `docker` group
- ! **VPN on?** Allow local network traffic
- ! **Restarted regtest?** Delete your wallet data too

See you in Turin

Agentic Dollars on Bitcoin · 17–18 October 2026 · BLOX, Turin

Get started
rgb.info/build/get-started

Documentation
docs.rgb.info

Tools to build
github.com/RGB-Tools

Protocol libraries
github.com/rgb-protocol

RGB Hub